Recommendations That Adapt to the Shopping Session

Balancing relevance and product variety in session-based re-ranking

TEAM G2
NGUYEN DINH NAM R ARULKUMARAN HO CHI THANH

2. Abstract

Project goal

Session-Adaptive Re-ranking

- Adapt the relevance–diversity balance as a shopping session evolves.

-> When should a re-ranker prioritise diversity over relevance?

Progress this sprint

- **Dataset:** Diginetica
- **Finding:** 90.2% of candidate lists contain one category
- **Decision:** investigate product-name diversity
- **Scope:** re-ranking + baselines + metrics

3. Motivation

The shopper's problem

A shopper browses a jacket, a watch and a backpack, then keeps comparing two running shoes. Early on, variety may help them explore. Later, close alternatives may help them decide.

Always similar

Narrows the shopper's options too early.

Always varied

Distracts a shopper who has a specific goal.

Why it is worth our effort

62.28% of queries are anonymous

Measured in our own audit of the Diginetica data: most query rows have no known user, so the current session is often the only signal.

The trade-off is still unsettled

Relevance and diversity can improve together or conflict, depending on the method and dataset.

Learning opportunities

We practice learning-to-rank, diversity re-ranking and careful offline evaluation on a real e-commerce dataset.

4. Task Statement

For each query, re-rank the supplied candidate products using only the session observed so far, choosing per session how strongly to trade relevance for variety.

Inputs (What the model sees)

- Earlier views, clicks and queries in the session
- Current queries or category
- The 1–200 supplied candidates, with category, price and names

Output (What the model produces)

- Top-K list balancing relevance and within-list variety

Evaluation target

- Products clicked after the current query

5. Proposed Method (1 of 2)

STEP 1

Summarise the session

Only events observed before the current query.

STEP 2

Score relevance

Popularity, original position, query–product overlap, session history.

STEP 3

Choose λ(s)

Use observed session diversity to determine λ(s)

STEP 4

Re-rank greedily

Start from ranked list S; move the best item into R until R holds K items [3].

RE-RANKING OBJECTIVE

$$f(i, R) = \boldsymbol{\lambda(s)} \cdot \text{relevance}(i) + (1 - \boldsymbol{\lambda(s)}) \cdot \text{diversity}(i, R)$$

relevance(i)	Relevance score of product i from Step 2, given the session so far
diversity(i, R)	Average token distance between product i and the products already in R
λ(s)	Balancing weight on relevance; [3] fixes λ = 0.5, we choose it per session
Scaling	Both terms min-max normalized per query (our addition); λ searched near 1 [6]

5. Proposed Method (2 of 2)

Building the four steps, by milestone

Slide 5 step	Deliverable	Milestone
1 · Summarize the session	Benchmark: history cut at query time, date split, scorer, token ILD	1
2 · Score relevance	Popularity, SERP order, token overlap; Session-kNN and GRU4Rec	2
4 · Re-rank greedily	Fixed-λ greedy re-ranking [3] and relevance–diversity curves	2
3 · Choose λ(s)	Threshold rule [3] and adaptive λ(s) from session diversity	3
3 + 4 together	Fixed vs adaptive comparison + ablations	3
Whole pipeline	Held-out results, confidence intervals and error analysis	4

Optional: learned λ(s) from several session features · a second similarity measure · purchase relevance and novelty

What we will compare

Q1 · MILESTONE 3

Does adapting λ to the session beat one tuned fixed λ and the threshold rule of [3]?

Q2 · MILESTONE 4

Which sessions gain: short, long or no history, and narrow or varied candidate sets?

Q3 · MILESTONE 4

How much variety does the relevance-only ranker already give on its own [6]?

6. Progress (1 of 2)

Work item	Owner	Status
Dataset profiling: sessions, queries, candidate sets and interaction statistics	Thanh	Done
Related work & task framing: session-based recommendation, relevance–diversity trade-off and re-ranking task	Nam	Done
Diversity suitability audit: category coverage and alternative product representations	Arul	Done
Benchmark setup: event-order audit + chronological train/validation/test split	Nam	In Progress
Evaluation pipeline: implement NDCG@K + Recall@K	Thanh	In Progress
Diversity evaluation: validate product similarity representation + ILD@K	Arul	In Progress

Team self-assessment

Satisfactory. Data understanding is complete

Link to the research question

The audit reshaped the question: variety now means within-category product-token variety.

6. Progress (2 of 2)

Technical insights

- 832,939 of 923,127 candidate lists (90.2%) hold one category, so category diversity cannot change.
- Median list has 197 candidates: about 134M query–item rows, so we need chunked processing.
- The bundled baseline's comment says weights 1/2/3; the code uses 0/1/2.
- timeframe units are undocumented; eventdate is the safe split field.

Fit with our goals

- Anonymous traffic suits a session-only method.
- Supplied candidates let us study balancing without building retrieval.
- Greedy re-ranking [3] keeps the relevance–diversity interaction easy to inspect.
- Session kNN is a strong, simple baseline to beat [3, 4].

Hazards and uncertainties

- Only about 37% of queries have an earlier view in the session (provisional), so we need a fallback policy.
- Event ordering must be validated before building prefixes.
- Memory and compute for 134M rows on [our hardware].
- Offline results cannot show satisfaction or sales.

7. Proposed Evaluation (1 of 2)

Evaluation Metrics

Metric	What it measures	Why we use it
NDCG@10, @20	Ranking quality / relevance	Primary metric; rewards relevant clicked/purchased products appearing higher
Recall@10, @20	Relevant-item retrieval	Secondary relevance metric; also used by [3]
ILD@10, @20	Within-list diversity	Primary diversity metric; directly follows [3]
Inference time / query	Computational cost	Practicality of session-time re-ranking; [3] also analyses runtime

7. Proposed Evaluation (2 of 2)

Comparison	Question it answers
Original order, popularity, S-POP [6]	Do simple signals already explain the results?
Session k-nearest neighbors [4]	Are similar past sessions enough?
Relevance-only ranker	How good is ranking without variety?
+ fixed-λ re-ranking [3]	What does one globally tuned λ achieve?
+ threshold rule [3]	Is a published session-diversity rule enough?
+ learned λ(s) (ours)	Does learning the balance add value?

8. Resources

1

Data

- Diginetica, CIKM Cup 2016 Track 2 [2]
- 923,127 queries · 1,127,764 clicks · 1,235,380 views · 18,025 purchases
- 184,047 products in 1,217 categories, about 901 MiB of CSV
- No extra data collection planned

2

Compute

- Own laptops / SoC Compute Cluster
- Python scripts plus notebooks for analysis

3

Software

- Python, pandas, NumPy, PyTorch
- Our own code: session features, greedy re-ranker, controller, evaluation harness (maybe referenced from [3])
- Fixed splits and run scripts for replicability

9. Schedule / Role Assignment

Milestone	Weeks	Task	Owner	Status
0. Dataset profiling & suitability audit	Wk 6	Profile Diginetica sessions, queries, candidate sets and interaction statistics	Thanh	Met
		Review session-based recommendation and relevance–diversity literature	Nam	Met
		Audit diversity signals: category coverage and product-token suitability	Arul	Met
1. Benchmark	Wk 7–8	Event-order audit + chronological train/validation/test split	Nam	In Progress
		Implement evaluation scorer: NDCG@K + Recall@K	Thanh	In Progress
		Validate product similarity + ILD@K	Arul	In Progress
2. Baselines	Wk 8–9	Original SERP + popularity baselines	Thanh	Planned
		Session-kNN + GRU4Rec relevance models	Nam	Planned
		Fixed-λ greedy re-ranking [3] + relevance–diversity curves	Arul	Planned
3. Adaptation	Wk 9–10	Fixed vs adaptive experiment + ablation protocol	Thanh	Planned
		Integrate adaptive re-ranking with relevance model	Nam	Planned
		Session-diversity signal + adaptive balancing policy $\lambda(s)$	Arul	Planned
4. Results	Wk 10–11	Held-out evaluation + confidence intervals	Thanh	Planned
		Model comparison + technical error analysis	Nam	Planned
		Session-level error + ethical/stakeholder analysis	Arul	Planned

10. Acknowledgements

DIGINETICA and the CIKM Cup 2016 organizers, for releasing the dataset and evaluation files.

Alexander Laktionov and Vladislav Grozin, authors of the sample baseline bundled with the competition data that we mainly referenced as inspiration.

References

[1] Carbonell, J. and Goldstein, J. (1998) 'The use of MMR, diversity-based reranking for reordering documents and producing summaries', in Proceedings of SIGIR '98. New York: ACM, pp. 335–336.

[2] DIGINETICA (2016) CIKM Cup 2016 Track 2: Personalized E-Commerce Search Challenge [Online]. Available at: https://competitions.codalab.org/competitions/11161 (Accessed: 18/09).

[3] Gharahighehi, A. and Vens, C. (2021) 'Personalizing diversity versus accuracy in session-based recommender systems', SN Computer Science, 2, Article 39. doi:10.1007/s42979-020-00399-2.

[4] Ludewig, M., Mauro, N., Latifi, S. and Jannach, D. (2019) 'Empirical analysis of session-based recommendation algorithms', arXiv:1910.12781.

[5] Phan, M.C., Tay, Y. and Pham, T.-A.N. (2017) 'Ensemble methods for personalized e-commerce search challenge at CIKM Cup 2016', arXiv:1708.04479.

[6] Yin, Q., Fang, H., Sun, Z. and Ong, Y.-S. (2023) 'Understanding diversity in session-based recommendation', ACM Transactions on Information Systems, 42(1), Article 24. doi:10.1145/3600226.